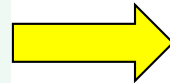


# Why Assignments Are Too Easy?

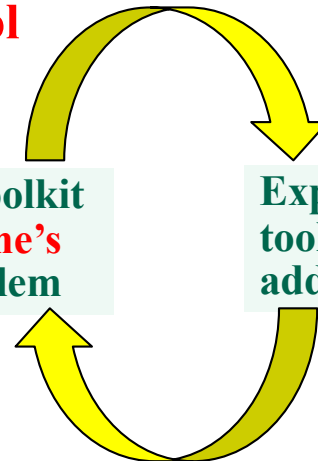
**A:** It's intentional. To provide hands-on seeds, so as to start using tools (MPI, OpenMP, CUDA, SYCL, *etc.*, or other mathematical & engineering tools at large) in a self-sustained use-learn cycle

**Most effective way to learn a new tool**

Understand  
the smallest set  
of essential  
mechanisms  
by reading a  
**minimalist  
program**

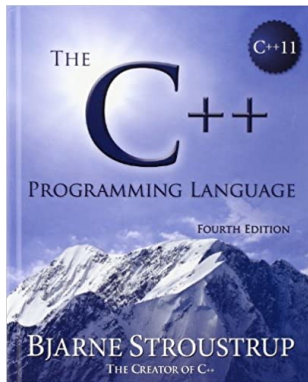


Use the toolkit  
to solve **one's  
own** problem



Expand the  
toolkit by  
adding

**Self-sustained  
Use-learn cycle**



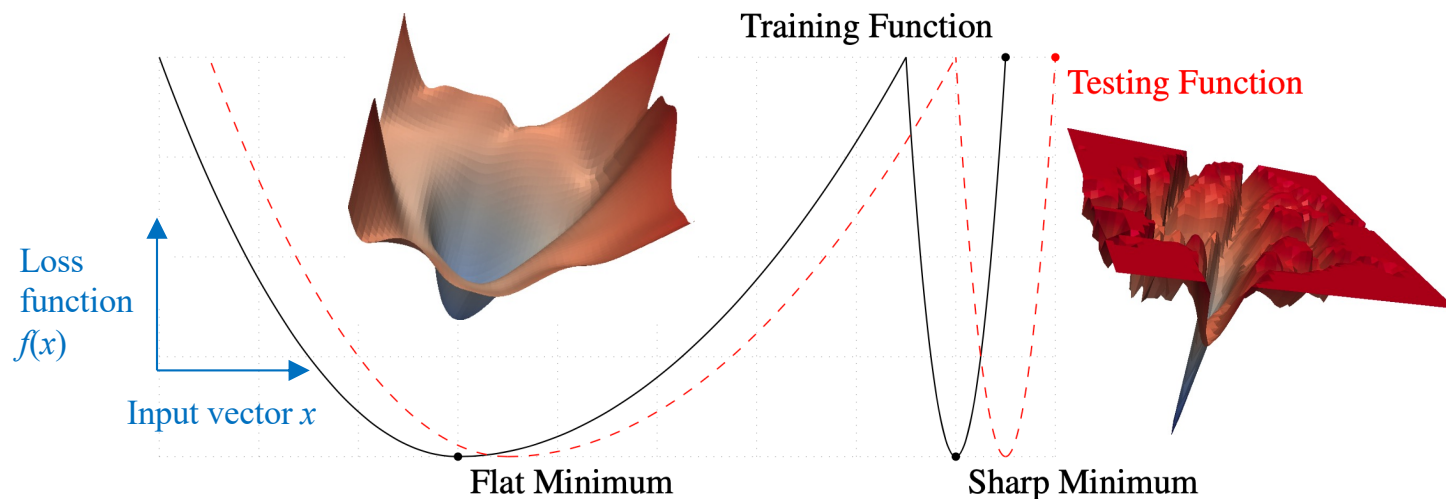
**Start the cycle in your final project**

**Alternative way**

“You bought a 1300+ page (480K+ word) book for \$40 or \$60, not a subscription service. How much support do you expect?”

# Learning-Theory Foundation

- **Minimum description length (MDL) theory:** Models described by fewer bits (*i.e.*, low Kolmogorov complexity) are more **generalizable**
- **Spectral theory:** Low-bit (low-complexity) models have **flatter loss landscape**



Keskar *et al.*, ICLR 2017 [<https://arxiv.org/abs/1609.04836>]

Forest *et al.*, ICLR 2021 [<https://arxiv.org/abs/2010.01412>]



**Allegro-Legato**

H. Ibayashi *et al.*, *ISC23—LNCS 13948*, 223 ('23); *arXiv*: 2303.08169

<https://github.com/mir-group/allegro>